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**SPECIFICATIONS FOR FABRICATION AND SUPPLY OF TURNTABLE LADDER 55
METER HEIGHT FOR FIRE FIGHTING AND RESCUE OPERATIONS.**

The Directorate of Maharashtra Fire Service receives specification for various fire tenders and special appliances from different Urban Local Bodies. Mostly, these specs have different compositions such as capacity of Water Tank, Gross Vehicle Weight, Accessories and Various Ancillary Equipment and thus in order to bring the uniformity for all these specifications across Maharashtra, following specification is designed and suggested by Directorate of Maharashtra Fire Services. The aforesaid specification is for "Turn Table Ladder (TTL) – 55 Mtrs". Details of specification are as follows.

The said TTL-55 Mtrs is mainly designed to cater to the needs of emergency that may arise due to fire and shall be need to carry out fire fighting and rescue operation. It is expected that ULB's shall have TTL-55 Mtrs to meet out fire related incidents and thus such type of vehicle shall be procured only after having fire tender / foam tender or quick response vehicle as well as the required manpower having proper training in the fire field from Maharashtra Fire Service Academy or National Fire Service College, Nagpur as the said vehicle is specialized vehicle and are usually imported one and required to especially carry out fire fighting as "fire tower" and "lifer" to carry out the rescue operation.

The Chassis shall be homologated from the appropriate authority in India in case not already an approved model or shall be supplied with COP issued by approved testing agency. The successful tenderer will have to complete the work as per specifications stipulated below and complete the vehicle in all respect to put into operation and ready to use till the time of delivery. The vehicle shall be ready to use and all the equipment and accessories shall be provided as per the provisions of Motor Vehicle Act 1988 and Central Motor Vehicle Rules 1989 and as per amendment time to time. The vehicle shall be guaranteed for a period of 24 calendar months from the date of supply of vehicle



including chassis and there after 3 years Comprehensive Annual Maintenance Contract (CAMC)

No part of the chassis shall be cut or mutilated by the successful tenderer without the prior permission of Chassis manufacturer. The successful tenderer will be solely responsible for the safe custody and proper maintenance of the chassis or any part thereof till the fabrication is completed and the vehicle is handed over to the ULB i.e. Municipal Corporation or Municipal Council with satisfactory test. The successful tenderer will have to complete the work as per specifications stipulated below and complete the vehicle in all respect to put into operation and ready to use.

Note:

1. Wherever makes of any equipment is given it shall always be read in continuation word "or equivalent"
2. Wherever the numerical is used indicating dimensions of any equipment or material, tolerance of +/- 10% shall be accepted. Except for working height and cage load.
3. Wherever the items / equipment is mentioned having NFPA or EN requirement, for all those equipment, proper certificate regarding the same shall be supplied by the OEM / Fabricator

SPECIFICATIONS FOR FABRICATION AND SUPPLY OF TURNTABLE LADDER 55 METER HEIGHT FOR FIRE FIGHTING AND RESCUE OPERATIONS.

1. GENERAL REQUIREMENT

- 1.1 The Turntable Ladder shall be designed specifically for the purpose of fire fighting and rescue to enable firemen to go up and down. It shall comprise of ladder sections with a cage mounted at the end of top ladder section and a lift for going up and down to rescue people. The entire unit shall be mounted on a Turn-Table on a Heavy Duty powerful Diesel - Engine chassis of VOLVO / MERCEDES BENZ / SCANIA make, 6x4 having approx. 5100 mm Wheel Base with fully factory built cabin and suitable capacity PTO. The Vehicle Chassis shall conform to EURO VI emission norms.
- 1.2 The Turntable Ladder shall be designed as per the designed, operational stability and structural strength as per the EN standard and the certificate to that effect issued by the competent agency shall be enclosed with the tender. The manufacturer must have built and supplied at least 50 Turn Table Ladder to various fire brigades in the world in the last 5 years, out of which 50% must be of the required or higher heights. The manufacturer must have also built and supplied minimum 5 TTL of same or higher height for fire and rescue operations in India in the last 5 years. Performance certificates of at least 3 such vehicles along with purchase order to be submitted with the offer.
- 1.3 The TTL shall be capable of use at any angle of elevation without any reduction of load capacity of the cage. It shall also rotate 360 degree at any angle of elevation as well as below ground level subject to ladder remaining clear of vehicle body.



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- 1.4 The appliance shall be compact and fast on the road and easily maneuverable in the crowded streets and around sharp corners. The overall dimensions shall not exceed the limits specified herein.
- 1.5 The height of the Turntable Ladder when fully extended shall not be less than 53 mtrs to the cage bottom from the ground.
- 1.6 The Turntable Ladder shall be micro processor based electro hydraulically controlled, permitting precise and easy operations under the most difficult conditions, with ample reserve of strength and stability.
- 1.7 Full safety interlocks shall be incorporated in the design so as to ensure complete safety in operations and long years of reliable and trouble free service, as far as possible the system shall be fail proof.
- 1.8 The design of the TTL shall allow a very large safety margin for extreme operating and climatic conditions. The safe working loads ratings shall include an allowance for the weight of water system and the reaction from the monitor jet while operation.
- 1.9 The Vehicle shall have a leveling system to adjust an angle not less than 7 degree and it shall be in automatic in nature with the manual override in case of emergency.
- 1.10 There shall be a full back up system for all ladder movements and outrigger movement in case of failure of main system. This shall be achieved with the help of electro-hydraulic pump (220 or 380 Volt) powered by the vehicle generator.
- 1.11 The Complete Movement of the TTL shall be computer controlled and stabilized and the system shall be checked for interference sensitivity according to standards.
- 1.12 The Control system of the TTL shall be fully tropicalised and able to operate in the temperature range up to +55 degree centigrade and in a dusty and Humid condition without reducing the maximum operating limits.
- 1.13 The elevation of ladder shall be possible from $-10 + 75$ degree.

2. TRUCK CHASSIS:

- 2.1 The Chassis shall be either VOLVO / MERCEDEZ BENZ / SCANIA make, 6x4 having approx. 5100 mm Wheel Base with fully factory built cabin and suitable capacity PTO. The Vehicle Chassis shall be a Right Hand Drive and shall conform to EURO VI emission norms. The Chassis shall be homologated from the appropriate authority in India in case not already an approved model or shall be supplied with COP issued by approved testing agency.
- 2.2 The engine shall be six cylinder, inline/'V' type, Diesel with direct injection, turbo charged with intercooler.
- 2.3 The engine shall develop minimum 370 HP.
- 2.4 The gearbox shall be **automated type** with the arrangement of suitable capacity PTO required for driving the hydraulic pump for ladder movements.
- 2.5 Rear Axle shall be Tandem Banjo type with Hub reduction and differential lock between the wheels and axles.
- 2.6 Chassis frame shall be 'C' Channel section made of high strength steel with cross members.



- 2.7 The Steering shall be integral power steering with fully adjustable steering wheel and column.
- 2.8 The Front and the rear Suspension shall be leaf spring type.
- 2.9 The Brakes shall be dual circuit airbrakes, disc type at front and drum type a rear axles with parking brakes acting on rear wheels.
- 2.10 Fuel Tank - Capacity shall be min 300ltrs with lockable fuel cap.
- 2.11 The Chassis shall be provided with 10 Tyres – plus spare tyre of suitable size as per GVW of vehicle.
- 2.12 The chassis shall be provided with single cab with RED RAL 3000 colour, made from high strength steel fully trimmed, external panels hot dip galvanized with hydraulic cab tilting mechanism. The cab shall be provided with adequate ventilation, rear view mirrors, windscreen glasses and windows, adjustable driver seat, wiper system and along with all other standard fitments.
- 2.13 Cab shall meet crash test requirement of ECE R29 norms.
- 2.14 The Electrical system shall be 24V, with suitable capacity batteries & Alternator for charging the batteries.
- 2.15 The chassis shall be supplied with standard tool kit, hydraulic jack, operator & workshop manuals.
- 2.16 The Chassis shall be fitted with suitable capacity Power Take Off Unit to drive the hydraulic pump for ladder movements.

3. MAIN OPERATING REQUIREMENTS & DIMENSIONS:

- | | | |
|-----|---|----------------|
| 3.1 | Max working Height: | 55 mtrs |
| 3.2 | Height to working cage bottom: | 53 mtrs |
| 3.3 | Working outreach with 300kg cage load: | 18 mtrs |
| 3.4 | Rotation – Continuous: | 360 degree |
| 3.5 | Safe working load in the cage on hard level | |
| | Ground with dry monitor: | 270 kg Minimum |
| 3.6 | Safe working load with monitor in the cage | |
| | Delivery up to 2000 LPM: | 180Kg Minimum |
| 3.7 | Loading capacity of lifting eye at the base ladder section: | |
| | ...2000 Kg Minimum | |



- 3.8 Operations at maximum outreach with full working load permitted in wind speed up to: 12.0 Mtr/ Sec.
- 3.9 The unit will be suitable for test at the load: 405 kgs at hard level Ground
- 3.10 Operating time at full stroke for all operations: As per EN standard
- 3.11 Overall length in traveling position not more than: 12 mtrs
- 3.12 Overall width of the vehicle: 2.50 mtrs max
- 3.13 Overall Height in traveling position: 3.95 mtrs max
- 3.14 Maximum width of the vehicle when Jacks are fully extended on both sides not more than: 5.3mtrs
- 3.15 Gross Vehicle Weight, depending on chassis type: As per CMVR
- 3.16 Operational weight according to DIN 70020: as per CMVR

4. CONSTRUCTION:

- 4.1 The vehicle shall be robust in construction; materials used in construction shall be carefully selected for lightness and durability. Use of timber shall be restricted in bodywork and use of rubber shall be avoided as far as possible. Ferrous metal parts shall be treated for anti - corrosion by a method other than electro-plating.

5. LADDER SET:

- 5.1 The vehicle shall perform the following functions/ operations
- Elevation
 - Depression
 - Extension & housing
 - Rotation 360 degree continuously in either direction
- 5.2 All the operations shall be electro-hydraulically operated with the help of hydraulic cylinders, wire ropes, chain etc. The system shall be purpose built to provide smooth takeoff, variable speed range and smooth slowdown, as per the EN standard.



- 5.3 There shall be Five/six ladder Sections, with telescopic movement made from high grade, corrosion resistant steel and shall have minimum wind catching area. The ladder sections shall be extended and retracted telescopically and simultaneously. Wind sensor to measure the wind speed shall be installed at the ladder tip with warning signal sent to the main control stand when wind speed exceed safe operation conditions.
- 5.4 Based on the selected outrigger position and cage load, the system selects automatically the maximum outreaches to all directions. The system capacity shall enable various outreach curves for each direction. The size of each sector shall automatically be defined based on position of outriggers. The outreach control system shall practically be infinitely variable.
- 5.5 The lower ladder section shall be screwed to turret. The ladder sections shall be welded construction; welding method shall be of latest technology to provide high durability and extreme accuracy. For high strength and minimum flexing of the ladder sections only high tensile strength steels shall be used.
- 5.6 The ladder elevation and lowering shall be controlled by two hydraulic cylinders that both have their separate safety devices and can alone carry the entire load in case of failure of any one of the cylinders.
- 5.7 All ladder sections shall move in a synchronized way and there shall not be any intermediate jerks during extension / retraction. Automatic slowdown mechanism at the beginning of the movement as well as end of the movement shall be provided to all ladder movements. All the moving sections shall be fitted with adjustable guides/ rollers to provide smooth and accurate movement. Various maintenance points shall be located well at hand either outside the ladder or behind easily removable covers.
- 5.8 The extension and retraction wire ropes shall be so laid; it should not cause any hindrance for climbing to a fireman. The ladder rungs shall be Anti Skid design and extension and retraction cables shall be provided with tension adjustment mechanism.
- 5.9 The ladder shall be possible to elevate from -10 to $+75$ degree minimum angle. The rotation movement shall be continuous through 360 degree at all angle of elevation except for the cabin protection area. In driving condition, the ladder set shall be placed on the ladder head rest.



- 5.10 All the sliding sections shall have maintenance free Nylon/Steel rollers for sliding movements and means shall be provided for the lubrication of these rollers at an easily accessible position.
- 5.11 Hook On type additional ladder shall be provided for the access to main ladder assembly from the ground. A lifting eye shall be provided at the head of the main ladder section to lift the load .
- 5.12 An attachment system shall be provided at the top of ladder and in cage for fixing water monitor. The monitor attachment shall be quick connecting type without the use of bolts and nuts. The monitor can be kept at some other suitable place on the vehicle.
- 5.13 All main ladder movements shall be possible individually and simultaneously. While using simultaneous movements there shall not be reduction in the speed of ladder movements. The ladder leveling (plumbing) movement shall be automatic in nature.
- 5.14 All the ladder movements except rotation, just before reaching the limit shall slowed down automatically until rest. In addition to this, the ladder lowering rate shall proportionally be decrease with increasing ladder length in order to avoid excessive swinging of ladder tip when ladder movement stops.
- 5.15 The functional, ergonomically designed main operating control consol shall be provided on the left hand side of turntable, with suitable operator seat. The following controls shall be provided on main control.
- 5.15.1 Lever/joy stick for rotation and elevation and depression.
 - 5.15.2 Lever/Joy stick for extension and retraction.
 - 5.15.3 Foot pedal for oil pressure ON/OFF (Dead Man type switch)
 - 5.15.4 Button for engine Start/Stop.
 - 5.15.5 Button for ladder leveling (Plumbing) ON/OFF.
 - 5.15.6 Button for rungs in alignment.
 - 5.15.7 Button for One man/Two man/Three man selection.
 - 5.15.8 Button for lighting ON/OFF.
 - 5.15.9 Button for emergency stop.
 - 5.15.10 Volume control for main control stand speaker and speaker at ladder top.
 - 5.15.11 Microphone for intercom system.
 - 5.15.12 Lever for emergency operation ladder and jacks.



- 5.15.13 Graduated Arc with pendulum.
- 5.15.14 Button for automatic housing of ladder in the head rest.
- 5.15.15 Display screen.(LCD or TFT)
- 5.15.16 The ladder set shall be primed and painted for long life span, treated against rust and corrosion.

6. HYDRAULIC CYLINDERS:

- 6.1 The Hydraulic cylinders shall be double acting, fitted with lock valves so as to prevent ladder set, working cage from lowering or the outriggers from retracting in case of pipe or hose failure.
- 6.2 The cylinders shall be provided with automatic dampers to prevent the pressure shocks and shall dampen the movement when a mechanical stop is reached.
- 6.3 Retraction of the outriggers shall be automatically prevented as soon as the ladder set has been lifted up from their transport position by way of electrical OR Hydraulic interlock system.
- 6.4 The ladder elevation and lowering has to be controlled by two hydraulic cylinders that both have their separate safety devices and both can alone carry the entire load in case of failure of any one of the cylinders.
- 6.5 The piston rods of the outrigger cylinders shall be protected suitably in order to protect piston rod from damage caused by any external impacts.
- 6.6 Lifting of the ladder set from the transport position shall be prevented before the outriggers are in support position and there shall be a limiting circuit to prevent damage to the Drivers cabin when the ladder is not clear of the cabin.
- 6.7 All the movements shall be automatically limited in their extreme position and the working cage shall be prevented from working outside of the permitted working range in any position.
- 6.8 An emergency stop switch shall be provided on both control panels, which shall switch off the hydraulic pressure of all movements and shall stop the vehicle engine.



7. TURN-TABLE:

- 7.1 The turret shall be fully integrated steel structure containing center post, slip rings, etc duly fastened to the main frame by means of a slewing ring.
- 7.2 The rotation for the turntable shall be controlled by hydraulic motor with brakes.
- 7.3 The base control station shall be attached to the turntable so as to rotate with it and be accessible in all positions of the turntable.
- 7.4 The hydraulic distributor (center post) shall be mounted in the center of the turntable at an accessible position and shall carry the hydraulic pressure and return lines, electrical supply lines allowing continuous rotation in either direction.
- 7.5 The fasteners retaining turntable to the rotation mechanism shall be of proper grade and shall be torque properly.
- 7.6 The rotation gearbox fastener shall be of proper grade and torque with proper backlash.
- 7.7 Pins securing the hydraulic cylinders shall be properly installed and secured.
- 7.8 The hydraulic hoses, tubing's and connections provided in the turret shall be free from kinks, chaffing or leaks.

8. STABILISING / JACKING SYSTEM:

- 8.1 The jacking system shall consist of four outriggers hydraulically operated, mounted in their housings. Each jack shall be fitted with adjustable guides to provide smooth and accurate movement of the outrigger beam. The outrigger piston rods shall be completely protected by closed steel profile.
- 8.2 The jacks shall be preferable diagonal configuration type; each side shall have two separate stabilizers.
- 8.3 The jack shall be provided with ground pressure sensors, which shall be correctly actuated before the ladder set is operated to ensure proper stabilization.
- 8.4 Each jack shall be provided with self-aligning footplate to spread the load evenly and allow the operation on uneven ground.
- 8.5 The system shall be able to level the vehicle up to 7 degree trough the 360° rotation with automatic leveling system.



8.6 The jacking system shall be controlled by two separate control panels provided on left and right side of the vehicle at rear with following controls.

- 8.6.1 Left side front and rear outrigger beam out.
- 8.6.2 Right side front and rear outrigger beam out.
- 8.6.3 Left side jacks down
- 8.6.4 Right side jacks down.
- 8.6.5 Automatic axle locking mechanism.
- 8.6.6 Cage folding in isolation switch.
- 8.6.7 Vehicle leveling gauge
- 8.6.8 Emergency stop push button

8.7 All the jack movements shall be infinitely variable within the full jacking width.

8.8 The Jacking systems shall allow operating each jack individually and the jack projection shall be recognized by the controlling system and the maximum outreach shall automatically be calculated as per the jack width.

8.9 The jacks shall be controlled individually or in pair with lever /joystick and the control panel shall be situated in such a position that, the operator will have clear look to the right and left hand side while extending the jacks. The control panel shall be located at the rear side of the vehicle and shall be covered with protection cover.

8.10 Yellow Flashing warning lights shall be provided at the outer most point of the jacks to identify the position of the jacks during night operation.

8.11 Four wooden spreader plates shall be provided for the use, when the vehicle is to be operated on soft ground.

8.12 The stabilizing system shall also preferably incorporate axle-locking mechanism.

8.13 The locker containing outrigger controls shall be fitted with an automatically operating door switch and a light for night operation.

9. **ELECTRONIC SAFETY AND OUTREACH SYSTEM:**

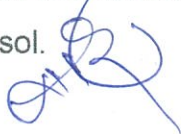
9.1 The computer-controlled and stabilized system shall allow the outriggers to be positioned and the system shall be capable to select automatically the maximum allowed outreach to front, rear, right and left side. Based on calculations and parameters saved in the system to guarantee exactly the same outreach regardless of the external influences like wind speed and direction, temperature, friction of the cylinders, etc.



- 9.2 The display units of the system shall show maximum possible outreach and position of the working cage in real-time along with other details.
- 9.3 The electronic system shall be made in CAN-BUS technology and approved according to the valid standards and directives. The system shall be EMC tested (EU directive 89/336/EEC) and CE type tested by TUV or any other appropriate agency.

10. CAGE & LIFT:

- 10.1 The working cage shall be attached to the tip of top section and it shall be foldable. There shall be sufficient space for three persons and safe working load shall not be less than 300 kgs. excluding the equipment permanently mounted in the cage. The leveling of the cage shall be controlled automatically by an electro/hydraulically mechanism with manual override in case of emergency. The controls for folding over and in working position shall be integrated into the control panel for the jacking system. The entrance to the cage shall be provided from the front and rear.
- 10.2 The cage shall be made of tubular steel/Aluminum profile, welded together and painted with special paint with high durability in case of steel cage. The dimensions of the working cage shall be such that it shall provide sufficient space for three men. The top railing shall be part of the cage door so that entering into the cage without bending is possible.
- 10.3 The cage shall be designed for 300 Kgs working load and shall be tested for min. 50% overload capacity.
- 10.4 The cage shall be quick removable type whenever needed.
- 10.5 An emergency cage levelling operation shall be provided in case of failure of electricity by means of Hand/Foot pump.
- 10.6 The cage control shall allow all ladder movement to be performed infinitely variable. During cage operation, the movement shall be automatically controlled by the safety system. When the ladder is operated from the cage, the speed of movements shall be the same as ladder control from main control consol.



- 10.7 The location of cage control shall be such that the operator shall have clear look on all side when operating the ladder from the cage. The Joy stick/Levers in the cage shall be identical that of the main console. The cage control panel shall always be visible and accessible, even when the additional equipment is used by the fireman. One xenon flood light shall be provided in front of the cage for illumination during night operations.
- 10.8 The cage control shall have following controls on the control panel.
- 10.8.1 Lever/joy stick for rotation and elevation and depression.
 - 10.8.2 Lever/Joy stick for extension and retraction.
 - 10.8.3 Foot pedal for oil pressure ON/OFF (Dead Man type switch)
 - 10.8.4 Button for engine Start/Stop.
 - 10.8.5 Button for rungs in alignment.
 - 10.8.6 Button for One man/Two man/Three man selection.
 - 10.8.7 Button for lighting ON/OFF.
 - 10.8.8 Button for emergency stop.
 - 10.8.9 Button for automatic housing of ladder.
 - 10.8.10 microphone-cum-speaker for intercom system.
 - 10.8.11 Operation display screen.(LCD)
 - 10.8.12 Button for cage light ON/OFF.
- 10.9 All ladder movements shall be stop automatically when the cage collusion system is activated.
- 10.10 A rescue lift shall be mounted at the rear of ladder set to be used for fast emergency rescue operation. Lift shall be use at any ladder rotation angle and also when cage is mounted at ladder tip.
- 10.11 Rescue lift capacity shall be not less than 300 Kg.
- 10.12 Lift shall be controlled by main control stand and wireless remote control and Access to the lift shall be possible also from the ground level at the rear or at left/right side.

11. HYDRAULIC SYSTEM:

- 11.1 The Hydraulic power shall be provided by a reliable and adequate capacity variable displacement axial piston pump, which shall be driven by the vehicle power take off.



- 11.2 When no operation of the ladder is activated, the pump shall rotate on minimum flow and minimum pressure. When the control lever or oil pressure foot paddle is pressed, the engine should automatically go to pre set rpm and also activate the necessary hydraulic oil supply to all the circuits.
- 11.3 There shall be a provision of instant couplings for attachment of manometer in each pressure line for checking pressure of each circuit.
- 11.4 The suitable filtration system of the hydraulic oil shall be provided on the vehicle to insure a good quality of oil entering into the system.
- 11.5 All hydraulic cylinders shall be double acting with hard chrome plated piston rods and shall be fastened by means of preferably self-aligning ball bearings to prevent lateral forces from damaging the seals or piston rods of the cylinders.
- 11.6 Hydraulic oil tank shall be provided into the turn table of suitably capacity. There shall be a proper heat dissipation system. The tank shall be fitted with oil level gauge, and suction connections with closing valves for easy maintenance and draining outlet with closing valve.

12. BACK-UP FOR THE HYDRAULIC SYSTEM:

- 12.1 The hydraulic power for all necessary ladder movements shall provide via a separate electro-hydraulic pump (220 or 380 Volt) . The electro-hydraulic pump shall be provided at suitable place for safe and easy operation.
- 12.2 There shall be a separate power generator mounted at suitable place, providing electricity to the hydraulic pump, which will provide independent means of hydraulic power in case of failure of main engine of vehicle. The power generator shall be able to start from main control panel.
- 12.3 In case of failure of main control panel system, emergency operation of all ladder movements shall be possible. All the ladder movements shall be controlled with hydraulic system from the main control consol.

13. CONTROLS AND SAFETY:

- 13.1 The Electrical supply needed for control system shall be taken from the vehicle battery which shall be charged when the engine is running.
- 13.2 When the vehicle is in operation yellow flashing warning lights mounted on the outriggers shall automatically remain on.



- 13.3 The engine starting and stopping switch shall be provided on all control panels and the engine speed shall be increased to the preset level as soon as the foot paddle is pressed.
- 13.4 All boom and rotation movements shall be controlled electro-hydraulically by means of proportional valves. The proportional valve shall not be sensitive to changes of ambient or oil temperature, and shall provide smooth, safe and very accurate movements even in most severe operating conditions.
- 13.5 The speed of the ladder for lowering and extension shall be automatically reduced at maximum outreach. The ladder lifting speed shall be reduced before the maximum elevation.
- 13.6 All control movements can be performed by the control system from both control panels and the outreach can be selected by the positioning the outriggers. The variable system shall consist of two displays, the graphical display and real time information about the outreach and the cage position and also to show possible movements according to cage position by animated arrows. In the text display there shall be main texts for:
- 13.6.1 Warnings
 - 13.6.2 Emergency situations
 - 13.6.3 Help manual
 - 13.6.4 Fault finding system
- 13.7 There shall be two independent drive elements to secured the ladder set of which each is capable of keeping the ladder in its present position during the ladder movements.
- 13.8 There shall be an inter lock, only permits the ladder sets to be lifted from the head rest and thereafter other movements are possible.
- 13.9 There shall be a load measuring system for stability and security of the ladder parts. The ladder movements shall stop automatically due to projection and / or additional load such as rescue persons, in conjunction with the jacking widths and the actual remaining loads.
- 13.10 An operating range display shall continuously show the total working range and the present load situation. The position of the ladder and the load limit in conjunction with the jacking widths shall display simultaneously.



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- 13.11 For the stability of the ladder, the speed of all the movements shall slow down smoothly and automatically before reaching the final position and automatically stop all movements in the defined final position.
- 13.12 For the tactical requirement during the emergency operation, there shall be a system which allows the ladder to reach the operational limit beyond the free standing limit, when the ladder head is supported against the wall.
- 13.13 When the ladder reaches to its operational limits. There shall be automatic final stopping of all ladder movements except for retraction and rotation.
- 13.14 There shall be a driver's cabin protection mechanism which will stop respective ladder movement to avoid collision of ladder with cabin.
- 13.15 There shall be an impact cut out which will stop all ladder movements.
- 13.16 An automatic hydraulically operated and electronically control leveling system (plumbing) shall be active all elevation and inclination angles. It shall guarantee that, the rounds are always horizontal up to angle of 7°.

14. OPEARTING RANGE DISPLAY:

- 14.1 An operating range display shall be provided at main control stand, and in the cage, which shall provide information to the operator. The various signals coming from the sensors, computer shall be processed and visually represented in the form of text or symbol on the display.
- 14.2 A scale down representation of the actual available range of operation shall be displayed showing exactly the ladder position, projection, ladder length, height and angle of elevation.
- 14.3 A microprocessor shall control and monitor all ladder movements and safety system, if the safe operating limit is reached, the ladder movement shall be blocked automatically and a corresponding massage / indicator shall be shown on display screen.
- 14.4 Display shall show text messages of operational faults including its consequences.
- 14.5 The operating range display shall be illuminated suitable for easy reading also during the night operation.



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15. CONTROLS AND INDICATORS IN DRIVERS CAB:

15.1 The following control and indicators shall be provided in driver's cabin.

- Visual warning for any of the equipment lockers being open
- Visual warning for the ladder set not being fully in transportation position
- Switch with visual indication for rotating beacons
- Switch with visual indication for siren unit
- Microphone for the public address system

16. BODY WORK AND EQUIPMENT LOCKERS:

- 16.1 The structure for the bodywork shall be made up of Aluminium profiles.
- 16.2 The complete external paneling of the rear body shall be made from Aluminium sheet fitted to the structural member either by gluing or riveting.
- 16.3 The complete flooring of the rear deck shall be made from non skid aluminum chequered plate of 3 mm thick properly riveted or bolted to the superstructure members.
- 16.4 For the easy access to the rear deck from ground level, there shall be sufficient numbers of recessed steps on both sides of the vehicle provided with suitable grab handles.
- 16.5 Sufficient number of lockers on each left/right side shall be provided for keeping various accessories and equipment's. The locker shall be so made that load distribution on both sides is equal. All the lockers shall be provided with roller shutters properly sealed for water and dust ingress. All the doors of the lockers shall be fitted with automatic switches activating the light as soon as the door is opened and also activating the warning light in Drivers cab.

17. THE WATER MONITOR:

- 17.1 Electrically operated monitor attached to the ladder head or in the rescue cage shall be provided with standard 63 mm. BIS male coupling.
- 17.2 The output of the monitor shall not be less than 1800 LPM at 10 bar and throw range shall be approx 40 mtrs.
- 17.3 The monitor can be moved vertically upward and downward up to -10° to $+70^{\circ}$. The horizontal movement up to 30° left side and right side



- 17.4 The monitor shall be fed with water through nylon delivery hose of 63 mm internal dia. meter of 55mtrs. in length with suitable male and female couplings.

18. INTER COMMUNICATION SYSTEM:

- 18.1 There shall be fully transistorized talk back inter communication system fitted between turntable and the cage.
- 18.2 The system shall be combined microphone and loudspeaker for hands free operation and shall be located in the cage. The turntable control station is also equipped with microphone, which shall be integrated in the loudspeaker with volume control.
- 18.3 The microphone and the loudspeaker shall be sealed properly and it shall be protected from the ingress of water, dust and humidity.

19. ELECTRIC SYSTEM:

- 19.1 The electrical supply shall be taken from the vehicle batteries, which are kept charged when the engine is running. Voltage of the system shall be 24 V DC and all circuits shall be provided with specific fuses depending on the current consumption of that circuit.
- 19.2 When the P.T.O. is switched on, yellow flashing warning lights located at each outrigger and ladder set pivoting point shall automatically be switched on.
- 19.3 Two rotating beacon lights shall be provided on each side of the drivers cab roof with multi-colour lens i.e. Red/Blue/white. The switch for switching the beacons on and off with suitable signal light shall be integrated in the control panel of siren cum public address system. The beacon light shall be either Bosch or Hella make or federal signal make.

20. SIREN AND PUBLIC ADDRESS SYSTEM:

- 20.1 There shall be an electric siren unit fitted on the roof of the vehicle cabin or at a suitable place with the control unit mounted conveniently inside the driver cabin. It shall have the fast (yelp) and slow (wail) sound modes with integrated switch for rotating beacon lights.



[Handwritten signature]

20.2 Command microphone, which is fitted with push-to-talk switch, allows the public address message to override the siren function. Operations are controlled by non glare-illuminated push buttons of different colors on control panel shall be provided.

21. PAINTING:

- 21.1 Before painting all surfaces of steel structures shall be carefully shot blasted after which they shall be primed and then applied the coat of approved paint. All the ladder set shall also be painted from all the sides.
- 21.2 For very high corrosion resistance of hollow structures such as steel profiles of the working cage, ladder set, outrigger beams and housings shall be treated with anti-corrosion protection preferably with "TECTYL".
- 21.3 The following Paint shades shall be used:
- | | |
|---|-----------------------|
| 21.3.1 Working cage & ladder sets: | Light grey RAL 7035 |
| 21.3.2 Ladder support, Turntable and related Cylinders: | Red RAL 3000 |
| 21.3.3 Body work including cabin: | Red RAL 3000 |
| 21.3.4 Chassis frame touch-ups: | Chassis original tone |

21.4 Fire brigade name and logo shall be provided in suitable positions on the vehicle.

22. ACCESSORIES:

- | | |
|--|----------|
| 22.1 Wooden outrigger ground pads/ plates with brackets | - 4 Nos |
| 22.2 Working range diagrams, at turntable & in the cage | - 2 No |
| 22.3 Marking of safe working load in the cage | - 1 No |
| 22.4 Unit type marked on the ladder. | - 2 Nos |
| 22.5 Warning labels and instruction plates | - 1 set |
| 22.6 Operation and maintenance manuals for TTL and Chassis. | - 2 sets |
| 22.7 Plug for 220 V power supply at the working cage | - 2 No |
| 22.8 Plug for 220 V power supply at the working cage | - 1 No. |
| 22.9 Lifting hook under the first ladder set of capacity 2000 kg | - 1 No |
| 22.10 Swiveling stretcher support c/w suitable folding stretcher to be installed on the cage when needed | - 1 No |



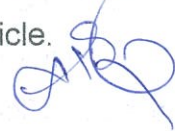
- 22.11 Positive pressure ventilator of at least 18000m³/h capacity C/w support for mounting on cage multifunction column - 1 No
- 22.12 Power generator 8 Kva capacity 220 Volt - 1 No
- 22.13 Delivery hose diam 2 ½" with STORZ/BSS coupling 55m length - 2 No
- 22.14 Set of tools & accessories required for the repairs & maintenance of TTL, chassis, & other systems - 1 set

23.DOCUMENTS :

- Following Documents has to be submitted during the bidding process and after the delivery such as
- General layout of the Turn Table Ladder
- User Manual & Instruction Booklet
- Necessary Certificate and Documents as mentioned in the tender

24.INSPECTION & TEST.

- 24.1 The Purchaser or its representatives shall have the right to inspect and/or to test the goods to confirm their conformity to the Contract. The Purchaser shall notify the Supplier in writing of the identity of any representatives retained for these purposes.
- 24.2 The inspection shall be carried out at manufacturer's factory prior to the delivery of TTL by Chief Fire Officer or its authorized representative. It is hereby suggested that there should be minimum three member panel in the inspection team. In absence of Chief Fire Officer, representative authorised by the Director, MFS shall carry out the inspection.
- 24.3 Cost of the inspection, minimum Railways AC- class II or Air_Fare to the nearest destination and local transport and lodging and boarding of optimum standards and shall be included in their offer. It is obligatory to the supplier to provide all the assistance and equipment for the inspection and testing of the vehicle.



- 24.4 The inspections and tests may be conducted on the premises of the Supplier or its subcontractor(s), at point of delivery and/or at the Good's final destination. When conducted on the premises of the Supplier or its subcontractor(s), all reasonable facilities and assistance including access to drawing and production data shall be furnished to the inspectors at no charge to the Purchaser.
- 24.5 Should any inspected or tested goods fail to conform to the specifications, the Purchaser may reject them and the Supplier shall either replace the rejected goods or make all alterations necessary to meet specification requirements free of cost to the Purchaser.
- 24.6 Acceptance Test :
- a) The Vendor, at the destination site, in the presence of authorized persons from the Purchaser, will conduct acceptance test of the goods delivered. The tests will involve trouble free operation of the goods under circumstances as envisaged by the Purchaser. There should not be any additional charges payable by the Purchaser for carrying out this acceptance test.
 - b) Within one months of the date of receipt of the machine (TTL) at the designated destination and after visual inspection, the purchaser will issue a taking over certificate in which the Purchaser will certify the date on which the vehicle has been so accepted. The taking over of vehicle found defective due to faulty design or workmanship or damage due to defective packing or otherwise not in conformity with the requirements of the Contract fails to be withheld until such time as the defects have been corrected.
 - c) If the Contractor fails to make good any defects, the Purchaser, may take at the cost of the Contractor such steps as may in all the circumstances be reasonable to make good defects. The issuance of taking over certificate shall in no way relieve the contractor of his responsibility for the satisfactory operation of the vehicle in terms of the specification.



25. TRAINING

- 25.1 The successful tenderer has to arrange training for the personnel of fire brigade department in handling, operation and maintenance of the above equipment. The training of minimum 4 sessions either at Fire Station of the concern ULB's or any other suitable location mutually agreeable to Head of the Fire Service of the ULB's and the contractor. The training shall cover operation, handling and maintenance of all the tools equipment and gears listed under this tender.
- 25.2 All the expenses towards the training shall be included in the cost in addition to training material and the cost of tools and equipment and consumable required at the time of training. The training program shall be chalked out in consultation with Head of the Fire Service or any other officer authorized by him.

26. MANUAL & INSTRUCTION BOOK:

- 26.1 Instruction books for the guidance of the user including both operation and normal maintenance shall be supplied for all the equipment in English language. The books shall include an item wise and illustrated spare parts list giving reference numbers of all the possibly wearing parts. The workshop manual and spare parts catalogue of chassis shall also be supplied with vehicle preferably with soft copy.

27. COMPREHENSIVE SERVICE MAINTENANCE CONTRACT (CSMC):

- 27.1 Comprehensive Service Maintenance Contract period will start from the date of expiry of 24 months warranty period .
- 27.2 The Contractor shall offer the vehicle with five years COMPREHENSIVE SERVICE MAINTENANCE CONTRACT which includes the cost of repairing of vehicle at periodic intervals or at the time of breakdown of vehicle including the supply of original spare parts.



- 27.3 The COMPREHENSIVE SERVICE MAINTENANCE CONTRACT shall be for superstructure as well as for the chassis. The servicing of the superstructure and the chassis shall be carried out strictly as per the manufactures recommendations at periodic intervals.
- 27.4 During the contract the vehicle shall be checked periodically at the interval of every three months and all the test and checks shall be carried out as per manufacturers recommendations.
- 27.5 The spare parts used at the time of periodical servicing shall be original and brand new.
- 27.6 Any break down of the vehicle shall be attended within 72 hrs. from the time of intimation of break down (telephonic / written) to the contractor.
- 27.7 The servicing and repairing of vehicle including chassis shall be carried out through skilled workers as certified by the manufacturer.
- 27.8 All the tools, consumables etc. required for the servicing of the vehicle shall be arranged by the contractor.
- 27.9 The servicing and repairing of the vehicle shall be carried out either at the fire station or at the fire service workshop or at the authorised workshop of vehicle manufacturer.
- 27.10 The complete servicing of the vehicle shall be carried out well in advance as per the provisions of Motor Vehicle Act and Central Motor Vehicle Rules when the vehicle is due for renewal of mechanical fitness certificate.
- 27.11 Any break down of vehicle on emergency call or on road shall be attended immediately.
- 27.12 The complete servicing and repairing of vehicle shall be carried out under the supervision of technical officer of department and all the instructions (oral or written) given by him time to time shall be incorporated / attended.
- 27.13 Any damage to the vehicle due to in proper handling or due to accident shall be attended promptly and the cost on account of such repairs including the cost of spare parts shall be got approved from Purchaser prior to such repairs.



27.14 Any dispute arise out of this contract, Purchaser will be the final authority and the decision given by him shall be binding to both the parties.

27.15 The tenderer shall give the details of work to be carried out at periodic interval of three months along with the offer.

27.16 The contractor shall maintain the log book of the vehicle and shall enter all the details of repairs /service of the vehicle carried out time to time and same shall be got certified either from Officer in charge of the fire station or from workshop in charge.



